

Please stick the barcode label here.

- (c) Given: mass of the metal block = 0.80 kg
heater voltage = 12 V
heater current = 4.0 A

- (i) By considering the maximum temperature rise of the metal block, calculate the specific heat capacity of the metal as found from the experiment. (2 marks)

Answers written in the margins will not be marked.

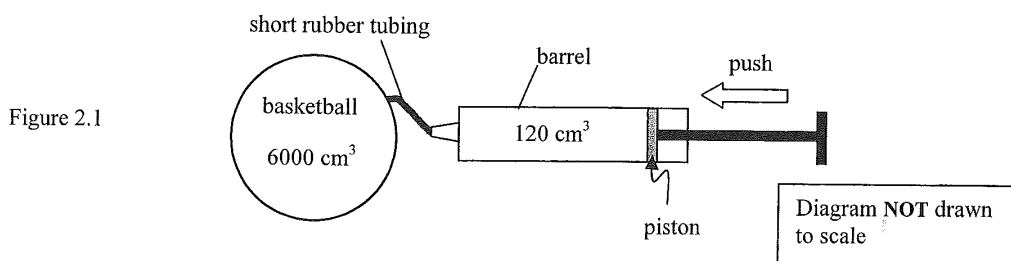
- (ii) Would your result be the same, higher, or lower than the actual value of the specific heat capacity of the metal ? Explain. (2 marks)

Answers written in the margins will not be marked.

- (d) This method is not suitable for measuring the specific heat capacity of a glass block. Explain. (1 mark)

Answers written in the margins will not be marked.

- *2. Figure 2.1 shows a basketball connected to an air pump via a short rubber tubing. By pushing the piston inward, the pump can compress 120 cm^3 of air inside its barrel at atmospheric pressure and room temperature into the basketball for each stroke.



Initially the volume of air inside the basketball is 6000 cm^3 and is at equilibrium with the atmospheric pressure 100 kPa . The basketball has to be pumped to a pressure of 156 kPa for an official match. Throughout the pumping process, the temperature of the basketball and the surroundings is assumed to be maintained at room temperature which is constant.

For each stroke, the valves of the pump (not shown in Figure 2.1) allow the air in the barrel to be completely pumped into the basketball and prevent it from going back into the barrel when the piston is pulled outward.

- (a) (i) Show that 3360 cm^3 of air, originally at atmospheric pressure, is required to be pumped into the basketball until its pressure is suitable for an official match. Assume that the volume of the basketball remains unchanged at 6000 cm^3 . (3 marks)

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- (ii) Hence estimate the minimum number of strokes needed to pump the basketball to the required pressure. (1 mark)

Answers written in the margins will not be marked.

(4 marks)

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(2 marks)

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- (ii) If the reading of the counter-weight's position on the beam carries an uncertainty of ± 0.1 cm, find the maximum error that corresponds to the result found in (a)(i). (2 marks)

- (i) Find the mass of the load. (2 marks)
- (a) A load is placed on the pan which is 5.0 cm from the pivot. The set-up is in equilibrium when the counter-weight of mass 50 g is 10.0 cm away from the pivot as shown.

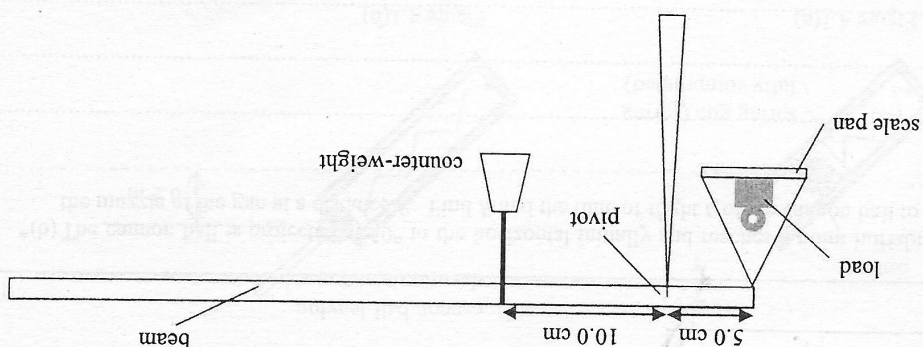


Figure 5.1

5. Figure 5.1 shows the use of a beam balance to measure the mass of a load. Without the load and the counter-weight, the beam with a scale pan at its left end is balanced and remains horizontal. ($g = 9.81 \text{ m s}^{-2}$)